

RSR week 4: T tests

1. An ecologist wants to measure average height of trees in a forest. She samples 100 trees and stores the variables shown. Which line gives the standard error of the sample mean (referring to image)

```
# 100 randomly sampled trees of one species
mean_height <- mean(trees$height)
sd_height    <- sd(trees$height)  # variation tree-to-tree
n_trees      <- length(trees$height) # 100
```

a) $\text{sd_height} / \sqrt{\text{n_trees}}$ ✓

b) $\text{sd_height} / \text{n_trees}$

c) $\text{sd_height} * \sqrt{\text{n_trees}}$

2. Two samples have similar SDs. Sample A has $n = 4,000$; sample B has $n = 200,000$. Which sample has the smaller SE of the mean?

a) Sample A, because smaller n gives a smaller SE.

b) **Sample B, because larger n gives a smaller SE.** ✓

c) Cannot tell without the t statistic.

d) They are the same; only SD affects SE.

3. For the patient data shown, the SD is 15. Which calculation gives the standard error of the mean blood pressure? Pay attention to the missings. (referring to image)

```
# A tibble: 10 × 2
  patient_id bp
  <chr>      <dbl>
1 P01      128
2 P02      142
3 P03      115
4 P04      NA
5 P05      156
6 P06      133
7 P07      149
8 P08      NA
9 P09      122
10 P10     147
```

a) $15 / \sqrt{10}$

b) **$15 / \sqrt{8}$** ✓

c) 15

d) $15 / 8$

4. You run `t.test(sbp ~ bmi_cat, data = whitehall)` to compare SBP between two BMI groups. What is the null hypothesis that we would disprove if the p value is below our chosen significance level?

a) **The mean SBP is equal in the two BMI groups.** ✓

b) The standard deviations are equal in the two groups.

c) The mean SBP is higher in the high-BMI group.

5. A 2,000-patient cold-relief trial gives the t-test output shown. The p value is ext
Which conclusion is correct?

```
> t.test(duration ~ group, data = trial)

Welch Two Sample t-test

data:  duration by group
t = -3.73, df = 1998, p-value = 1.96e-04
alternative hypothesis: true difference in means between
group drug and group placebo is not equal to 0
95 percent confidence interval:
 -0.153 -0.047
sample estimates:
mean in group drug mean in group placebo
        6.9         7.0
```

a) The CI proves the true difference is exactly -0.2 days.

b) The result is not significant so the drug likely has no effect.

c) The effect is statistically significant but may not be clinically meaningful.

